

Usability and Accessibility Test Report of VotingWorks VxScan, Version 4 with 39 participants for VVSG 2.0, Section 8.3

Report Based on ISO/IEC 25062:2006 Common Industry Format For Usability Test Reports

**VotingWorks VxSuite, Version 4, focused on VxScan with audio
and PAT jacks.**

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1.0 Executive Summary

A usability test of VxMark was conducted between April 15 and May 4, 2026 at various locations in the San Francisco Bay area by VotingWorks. The purpose of this test was to fulfill Section 8.3 of the Voluntary Voting System Guidelines (VVSG).

During the usability tests, a total of 39 testers used the VxMark in a simulated election. The election consisted of one test ballot with 12 contests, including:

- State and Local contests
- Partisan and nonpartisan contests
- Single member and multimember contests
- Ballot questions (retention races)
- Ballot propositions (reflecting referendum, ballot initiatives, and constitutional amendments)

This ballot used for the test is based on the test ballot developed by the National Institute of Standards and Technology (NIST), to provide opportunities for various voting tasks. The instructions includes 22 tasks and that model typical ballots from around the country, including:

- Voting for names at various locations within a list of names
- Voting a partial slate in a multimember contest
- Skipping elements of a ballot
- Changing a vote from the review screen
- Write-in votes
- Reviewing a ballot
- Casting a ballot

Following the conclusion of the testing, the results were analyzed to determine participants' effectiveness, efficiency and satisfaction using the VxMark.

During the usability test, the testing team collected and analyzed the following types of data:

- Number and percentage of ballots successfully submitted/completed (number and percentage)

- Percent of tasks completed without any errors
- Count of assists provided (number or percent of testers needing assists)
- Skipped contests
- Time to complete the voting session
- Rating of Audio
- Rating of Translations
- Rating of write in process
- Rating of changing contest process
- testers' confidence that they had used the system correctly
- testers' overall satisfaction with the system

The testing team also collected less structured feedback on the VxMark during, and after, voting sessions, which included general feedback on the voter's experience during the session and recommendations from the voter on improvements to the system.

Below is a high-level summary of the results:

Measure	Description	Usability Test Results
Successful completion	Average success rate of the 22 tasks performed by testers from the general population.	100.0%
Ballots cast without any assists	Count of the number of testers who were able to submit their ballot without any errors requiring assistance.	32 of 39 ballots (82.0%)
Count of assists provided	Number of testers and assists provided to testers during the usability test.	39 Assists given to 7 testers

VotingSession Time	Mean time taken per test participant to complete the process of activating, filing out and casting the ballot.	15 m 45 s
Rating of Audio quality	Number of audio users and percent who thought audio worked well	28 of 39 testers used this 92.9% thought the audio worked well
Rating of Translations	Number of audio users and percent who thought translations worked well	7 of 39 testers used this 100.0% thought the translations worked well
Rating of Write in Process	Mean rating of write in process based on 1-5 scale	3.47
Rating of Process to Change selections	Mean rating of change selection process based on 1-5 scale	4.05
Voter Confidence	Mean confidence level expressed by testers that they believed they voted correctly and the system successfully recorded their votes.	4.44 Confidence Level

Voter Satisfaction	Mean satisfaction level expressed by testers in response to a post-test satisfaction questionnaire focused on VxMark.	4.25 Satisfaction Level
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2.0 Introduction

2.1 Full Product Description

During the usability test the VxMark version 4.1 was evaluated. The VxMark version 4.1 was used for ballot marking and review. This version is the same as what will be provided to the Voting System Test Laboratory.

The VxMark is a tabletop ballot marking device. It allows all voters to make selections in various interaction modes, review their ballot, and print their ballot. The VxMark includes an accessible controller that can be used to navigate through the ballot if the voter is unable to interact with the touchscreen. Audio can be used via headphones, which provides instructions on how to use the accessible controller, navigate through the voting process, make ballots selections, and send the ballot for printing. The voter can cast their ballot on the VxMark, with tabulation occurring at a later point using VxScan or VxCentral. The voter may also take their VxMark ballot and cast it on the VxScan at the polling place.

To enable testers to test scanning the VxMark ballots on the VxScan, we had a VxScan and ballot box on location for all testing sessions and gave testers the option to cast on VxScan. The results of that testing are included in the VxSuite 4.0 certification campaign and are not the focus of this report.

2.2 Test Objectives

The usability test objectives include:

- To assess the effectiveness of the VxMark by measuring the abilities of users to successfully complete and submit a ballot.
- To assess the efficiency of the product by measuring the average time to complete a voting session.
- To assess the user satisfaction of the system by measuring average voter confidence.

- To gather feedback on the experience of using audio, translations, write ins, and changing ballot selections.
- To gather open ended, unstructured feedback on the system and voter experience.

3.0 Method

3.1 Participants

Participants were recruited by VotingWorks staff via in person outreach.

A total of 39 testers, with a varying mix of backgrounds and demographic characteristics were selected to participate in the usability test. All participants were over the age of 18, eligible to vote in the U.S., and able to communicate in English.

The usability tests used individuals whose visual acuity is less than 20/70 but greater than or equal to 20/200; the usability tests used individuals who can only read large-print, high contrast text; and used individuals who are visually impaired. Wearing corrective lenses did not assist these individuals in reading normal-sized text. VotingWorks had 12 testers with low vision who were able to complete the testing session, without assistance. More information available in the table below and Appendix A.

Please see Appendix A for a full spreadsheet of participant demographics.

Breakdown by demographics for 39 total participants

Demographic	All Participants
Sex	[12] Men
	[26] Women
	[1] Other

Age	[0] 18-24 years
	[7] 25-34 years
	[12] 35-44 years
	[8] 45-54 years
	[6] 55-69 years
	[6] 70+ years
Education	[6] Some high school
	[3] High school
	[11] Some college
	[16] College degree
	[3] Graduate/professional degree
Race/Ethnicity	[1] American Indian/Alaska Native
	[6] Asian
	[5] Black/African-American
	[10] Hispanic/Latino
	[1] Pacific Islander
	[16] White/Caucasian

Disability	[14] Blind
*Many testers identified more than one disability	[10] Low vision
	[1] Deaf
	[1] Hard of Hearing
	[9] Limited dexterity
	[1] Intellectual and Developmental disability
	[0] Limited or low literacy
	[2] Mobility
	[0] Brain/cognitive
	[6] NA
Language Used	[36] English
*several testers noted they were bilingual and chose to vote a ballot in another language	[5] Spanish
	[1] Chinese (Traditional)
	[1] Chinese (Simplified)
Geographic Distribution	[19] Urban
	[19] Suburban
	[1] Rural

Years of Voting Experience	[7] None/Chose not to answer
	[2] 2-5 years
	[3] 6-10 years
	[8] 11-20 years
	[19] 20+ years

Number of elections in the past 4 years	[8] None
	[6] 1-2
	[7] 3-5
	[18] 6+

3.2 Context of Use in the Test

3.2.1 Tasks

During the usability test, participants were instructed to vote in a simulated election consisting of one test ballot with 12 contests, including:

- State and Local contests
- Partisan and nonpartisan contests
- Single member and multimember contests
- Ballot questions (retention races)
- Ballot propositions (reflecting referendum, ballot initiatives, and constitutional amendments)

Our usability testing team based the test ballot on the medium complexity ballot created by NIST in order to cover all of the possible errors it incorporates.

Using this ballot, participants were asked to perform 22 tasks that were selected to model typical ballots from around the country, as well as to thoroughly test the voting system's capabilities and usability, including:

- Voting for names at various locations within a list of names
- Voting a partial slate in a multimember contest

- Skipping elements of a ballot
- Changing a vote from the review screen
- Write-in votes
- Reviewing a ballot
- Casting a ballot

Participants were provided an overview of the system, description of all tools and accessible options, and were asked to perform the task of marking, reviewing and casting a ballot without assistance. A task was considered successful if the participant was able to cast a vote in a way that matched their preference.

Data was collected for each task, including successful completions, time to complete a voting session, number of errors requiring assistance, number of assists provided, and voter confidence for each task.

Please see Appendix D for the test ballot.

3.2.2 Test Location

The VxMark is intended to be used at designated polling locations across the U.S., including schools, libraries, churches and other public facilities large enough to house multiple voting stations.

In order to simulate this environment, the test was conducted at several locations:

- Private residences for language testers
- Ability Now, Bay Area
- SF Lighthouse

3.2.3 Voting Environment

During an actual election, testers are expected to use the voting system provided at the polling location. Testers may have experience with a wide-range of systems or may only have experience with one type of system.

During the usability test, all participants were instructed to use VxMark as if this system was implemented at their local polling location. They were provided an overview of the equipment, accessible features, and then told to begin their voting session. The test administrator only intervened if assistance was requested by the voter.

3.2.3.1 Display Devices

The display is a 15" HD (1920 x 1080) touchscreen, specifically the Elo 1504LS. It can be tilted forward or backward by a voter based on their preferences. The screen includes an embedded speaker which is used for machine chimes and alerts. The screen rotates horizontally and then tips all the way back into the case for storage.

3.2.3.2 Audio Devices

The accessible controller sits to the right of the display, either in its cradle or beside VxMark outside of its case. It is the commercially available Storm AudioNav 9-Key Keypad. Four keys navigate directionally (up, down, left, right), one key is used to make selections, a pair of keys are used to control audio volume, and another pair of keys are used to control audio playback speed.

The headphones plug directly into the 3.5mm audio jack on the accessible controller. Any standard headphones can be used if a voter prefers, but note that the default volume levels were calibrated for the headphones that were provided with VxMark. Voting session audio plays through the headphones and only through the headphones. The volume and rate adjustments will have no impact on system sounds played over the speakers.

3.2.3.3 Input Devices

During the test, participants could use: headphones, jelly buttons, the accessible controller, and/or the touch screen. We also offered a stylus, if that made it easier to use the touchscreen.

3.2.4 Test Administrator Tools

During the usability test, various tools were used to facilitate the test sessions, including:

- Informed Consent (See Appendix B)
- Instructions for Participants (See Appendix E)
- Post-test Satisfaction Questionnaire (See Appendix C)

Participants' votes were recorded by the system, similar to a real-world election. Test facilitators recorded time by entering that information on the questionnaire. Start time (when the tester clicked the screen to bring up their ballot) and end time (when the voter cast their ballot) are recorded in the questionnaire and results.

3.3 Methodology

During the usability test, participants interacted with the VxMark and cast their ballot on VxScan. Each participant used the system set up in the same way and was provided with the same ballot, system orientation, and instructions. Each participant used the system set up in the same way and was provided with the same ballot, system orientation, and instructions.

The system was evaluated for effectiveness, efficiency and satisfaction. To evaluate these factors, the usability team collected data on:

- Number and percentage of ballots successfully submitted/completed number and percentage
- Percent of tasks completed without any errors
- Count of assists provided (number or percent of testers needing assists)
- Skipped contests
- Time to complete the voting session
- Rating of Audio
- Rating of Translations
- Rating of write in process
- Rating of changing contest process
- Testers' confidence that they had used VxScan correctly
- Testers' overall satisfaction with VxScan

Additional information about the various measures and associated metrics can be found in the section on Usability Metrics.

3.3.1 Procedure

Upon arrival, participants were greeted and asked to complete a Pre-Test Questionnaire (See Appendix C) to ensure that they qualified for the test. Participants who did not meet the qualifications were allowed to test and provide feedback, but their results are excluded from the data analyzed in this report.

Participants meeting the qualifications were asked to review and sign an Informed Consent (See Appendix B), which described their rights during the study and was provided in four languages: English, Spanish, Chinese (Traditional), and Chinese (Simplified). Participants were then provided an overview of the voting system and assistive technology options. Test administrators described the different settings that could be used by the tester and helped orient anyone using the accessible

controller, so they could feel the buttons and easily identify the help button on the controller.

After this overview, they were given the following instructions:

"Read this to tester right before you begin the test: "Please attempt to follow the instructions provided on the following screens and/or through audio provided. I will be here to provide basic assistance, similar to what a poll worker can provide in a polling location. Please feel free to share your feedback and comments out loud as you work through the ballot. If you are stuck and cannot continue, please inform me by raising your hand and/or verbally asking for help. I will be observing and taking notes from a few feet away."

During the usability test, test facilitators observed users' interactions and noted feedback they provided as they worked through the ballot. The test administrator only provided assistance when a user was stuck and asked a question. The assistance was limited to the question asked. Test administrators did not walk the voter through the ballot. Test administrators noted the start and end time of each session. When the participants finished the test, they were asked to complete a Post-Test Questionnaire (See Appendix C).

At the conclusion of the test, participants were thanked for their time and compensated with gifts cards for their time and energy.

Two staff members participated in this test.

3.3.2 Participant General Instructions

The participants were instructed that they would work alone during the test, but that they could ask for the same sort of assistance or the same sort of questions they might ask an election worker.

3.3.3 Participant Task Instructions

Participants were also provided with instructions on how to use different accessible features, reminded to read instructions and follow screen prompts, and to make sure they reviewed their selections on screen prior to printing their ballot. They were reminded and encouraged to write in candidates and review and change selections as they worked through the ballot. These instructions were provided to users orally.

See Appendix E for the instructions for participants.

3.4 Usability Metrics

The usability test collected various metrics for effectiveness, efficiency and satisfaction.

Name	Measure	Description
Effectiveness: Completion Rate	Ballots successfully submitted/completed	Percentage of test participants who were able to complete the process of voting and cast their ballots so that their ballot choices were recorded by the system.
Effectiveness: Errors	Tasks completed according to the on screen instructions without errors or assistance	Percentage of tasks that were completed without any errors. An error might involve a voter not following screen instructions to navigate through the ballot
Effectiveness: Assists	Number of participants needing assists	Count and percentage of the number of participants needing assists
Effectiveness: Assists	Count of assists provided	Count of the number of times assistance was given to participants.
Efficiency: Time on task	Average session time	Mean time taken per test participant to complete the process of activating, marking, and casting the ballot, both overall and by interaction mode

Satisfaction: Audio	Satisfaction with audio quality	Satisfaction expressed by testers in response to a post-test satisfaction questionnaire.
Satisfaction: Translations	Satisfaction with translation quality	Satisfaction expressed by testers in response to a post-test satisfaction questionnaire.
Satisfaction: Write in process	Satisfaction with write in process	Mean satisfaction level expressed by testers in response to a post-test satisfaction questionnaire
Satisfaction: Changing a selection	Satisfaction with changing a ballot selection	Mean satisfaction level expressed by testers in response to a post-test satisfaction questionnaire.
Satisfaction: Overall Satisfaction Rating	Average voter satisfaction	Mean satisfaction level expressed by testers in response to a post-test satisfaction questionnaire.
Satisfaction: Confidence	Average voter confidence	Mean confidence level expressed by testers that they believed they voted correctly and the system successfully recorded their votes.

4.0 Results

4.1 Data Analysis

To analyze the data, each session was scored for completeness, accuracy, errors, number of assists, and time to complete. In addition, the test team analyzed testers' satisfaction and confidence using various post-test questionnaires.

4.2 Presentation of the Results

This section details the performance results for effectiveness (completion rate, errors, assists), efficiency (time on task) and satisfaction (satisfaction and confidence rating).

Name	Measure	Results
Effectiveness: Completion Rate	Ballots successfully submitted/completed	Of the 39 participants included in the test, 39 were able to successfully cast their ballot.
Effectiveness: Errors	Tasks completed according to the instructions without any errors	32 of 39 (82.0%) testers were able to complete a voting session according to instructions without any assists
Effectiveness: Assists	Number of participants needing assists	7 [18.0%] of the participants needed assists
Effectiveness: Assists	Count of assists provided	A total of 39 assists were provided to those participants.
Efficiency: Time on task	Average session time	The average session time was 15 minutes 45 seconds.

Satisfaction: Audio	Satisfaction with audio quality	25 of 39 testers used audio during their session. All said it was satisfactory on VxScan. (Asked as a YES/NO question)
Satisfaction: Translations	Satisfaction with translation quality	6 of 39 testers used translations. All 6 testers thought the translations were satisfactory. (Asked as a YES/NO question)
Satisfaction: Write in process	Satisfaction with write in process	Average voter satisfaction was 3.47. Testers were asked to give their satisfaction rating on a scale of 1-5, with 1 being poor and 5 being excellent.
Satisfaction: Changing a selection	Satisfaction with changing a ballot selection	Average voter satisfaction was 4.05. Testers were asked to give their satisfaction rating on a scale of 1-5, with 1 being poor and 5 being excellent.
Satisfaction: Overall Satisfaction Rating	Average voter satisfaction	Average voter satisfaction using VxMark to cast a ballot was 4.25. Testers were asked to give their overall satisfaction rating on a scale of 1-5, with 1 being poor and 5 being excellent.

Satisfaction: Confidence	Average voter confidence	Average voter confidence was 4.44. Testers were asked to give their confidence rating on a scale of 1-5, with 1 being poor and 5 being excellent.
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More detailed results can be found in Appendix F.

5.0 Conclusion

Overall

Most testers expressed that they had a very positive experience.

Controller Challenges

There were many blind users who struggled to familiarize themselves with or use the controller. This was much more common in our sessions at the East Bay Center for the Blind than in our sessions at SF LightHouse. This may be because the SF LightHouse participants were more likely to have opted into the session intentionally and be more tech savvy, whereas the East Bay Center for the Blind participants were just approached at gatherings at the center.

Audio Flow Control

There was a lot of feedback around audio control. In particular, there was no way to repeat instructions.

If you missed the instructions at the beginning in particular, and the poll worker hands you the headphones, you're just plopped into silence. The audio and instructions for the controller start immediately after the poll worker removes the card. Ideally there should be a way for either the poll worker or the user to start the audio instructions

VotingWorks made updates to the software to resolve these issues prior to submitting VxSuite 4.1 for testing.

Accidental Undervotes

Many users undervoted multi-seat races unintentionally, only noticing in the review phase. In multi-seat races, after each selection, VxMark reads back the number of votes remaining as “Votes remaining in this contest: X”, and noted they are more familiar with the “Vote for up to X” language.

VotingWorks made updates to the software to resolve these issues prior to submitting VxSuite 4.1 for testing.

Proposition Information on Review Screen is Inadequate

Many users across groups found that, on the review screen, the title of propositions was not sufficient to review their choice. They may not remember what “Ballot Proposition 7” is about so, to truly confirm their choice, they have to navigate into that contest. Many suggested a preview.

VotingWorks made updates to the software to resolve these issues prior to submitting VxSuite 4.1 for testing.

Accidental Activation

Test administrators witnessed one tester report to them an instance of accidental activation of actions through an unintended gesture via the touchscreen or other accessible inputs.

VotingWorks made updates to the software to resolve these issues prior to submitting VxSuite 4.1 for testing.

Appendixes

- [Appendix A: Participant Demographics](#)
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Appendix A: Participant Demographics

 Copy of UAT Data Collection Form 2026 (Responses)

Appendix B: Informed Consent

 [Copy of Informed Consent Form - VxMark_VxScan UAT SF - 2026 - Chinese, sim...](#)

 [Copy of Informed Consent Form - VxMark_VxScan UAT SF - 2026 - Spanish.doc...](#)

 [Copy of Informed Consent Form - VxMark_VxScan UAT SF - 2026 - English.docx...](#)

 [Copy of Informed Consent Form - VxMark_VxScan UAT SF - 2026 - Chinese, tra...](#)

Appendix C:Test Session Questionnaire

 Copy of UAT Data Collection Form 2026 - Google Forms.pdf

Appendix D: Test Ballot Specification

 PROOF-official-precinct-ballot-Precinct_1-1_zh-Hant.pdf

 PROOF-official-precinct-ballot-Precinct_1-1_zh-Hans.pdf

 PROOF-official-precinct-ballot-Precinct_1-1_es-US.pdf

 PROOF-official-precinct-ballot-Precinct_1-1_en.pdf

Appendix E: Instructions for Participants

Prior to starting the session, test participants were given the following instructions orally after receiving an overview of the accessible options of the VxMark & VxScan:

1. Please let me know if you need headphones and/or tactile input.
2. Please use the settings menu to select the contrast , language, and font size that works best for you.
3. Please read the instructions at the beginning of each contest carefully and pay attention to screen alerts. Work through each contest selecting the candidates you choose.
4. Please complete a write in entry for a contest of your choosing or make changes in any contest or on the review screen prior to printing your ballot.
5. Please review all your selections after you've completed your ballot and make any corrections you believe are necessary.
6. Please press the cast ballot option to print your ballot.
7. After printing your ballot, we will proceed to the precinct scanner for you to cast your ballot. Please let me know when you are ready to cast your ballot.
 - a. At the scanner - Please let me know if you need headphones and/or tactile input for the precinct scanner. I will remind you of the settings options for VxScan when you are ready to cast your ballot.
8. The test administrator is available to answer questions as a poll worker, but will not otherwise assist you with your test session.
9. Please feel free to speak feedback or frustrations out loud, the test administrator will note that feedback for system improvements.
1. When you are ready, please select the button at the bottom of the screen. n.

Appendix F: Results

The results are available here:

 [Copy of UAT Data Collection Form 2026 \(Responses\)](#)